

Momentum and Profitability in European Small-Cap Equities*

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Abstract

This document summarizes the construction and evaluation of a cross-sectional equity strategy on European small caps (EUR 180M–1.8B market capitalization, 15 markets, 2007–2025), built on point-in-time Compustat Global data. A two-factor composite of 12–2 momentum and gross profitability (cf. Jegadeesh & Titman, 1993; Novy-Marx, 2013), held long-only in the top decile with monthly rebalancing, earns an annualized excess return of 8.9% over the equal-weighted universe out of sample, net of assumed trading costs (2020–2025; Newey–West $t = 4.04$, information ratio 1.31), against 9.0% ($t = 5.34$) in sample. A variant replacing annual gross profitability with quarterly income-before-extraordinary-items over assets earns 14.4% ($t = 4.24$) out of sample. A market-neutral implementation, shorting a European small-cap ETF against the long book with a rolling 24-month hedge ratio, attains an out-of-sample net Sharpe ratio of 1.4–1.7 for the quarterly variant (roughly 1.0 adjusted for return autocorrelation), with CAPM market beta statistically indistinguishable from zero. A capacity study replays the strategy under participation-limited execution and puts the useful ceiling near EUR 500M. Data-integrity safeguards, cost sensitivity, and methodological limitations are documented, including partial reuse of the holdout window across research iterations.

1 Overview

The project implements a monthly cross-sectional factor model for European small-cap equities: universe construction from point-in-time vendor data, factor computation under explicit publication-lag assumptions, a long-only decile backtest with an in-sample/out-of-sample split fixed in advance, and post-hoc diagnostics (factor-model alphas, country attribution, tail behavior). Two further sections address implementation: a market-neutral overlay that shorts a European small-cap ETF against the long book, and a capacity study that replays the strategy under an explicit execution model with daily participation limits. The emphasis throughout is on the mechanics that make backtest numbers trustworthy: look-ahead prevention, data-integrity checks, and test-enforced invariants. The factor signals themselves are deliberately taken from the published literature.

2 Data and universe

Sources. Security-level daily prices, shares outstanding, and listing metadata are from Compustat Global via WRDS. Annual and quarterly fundamentals come from `g_funda` and `g_fundq`. Foreign-exchange conversion uses ECB daily reference rates, and all returns are converted to EUR at month-end rates.

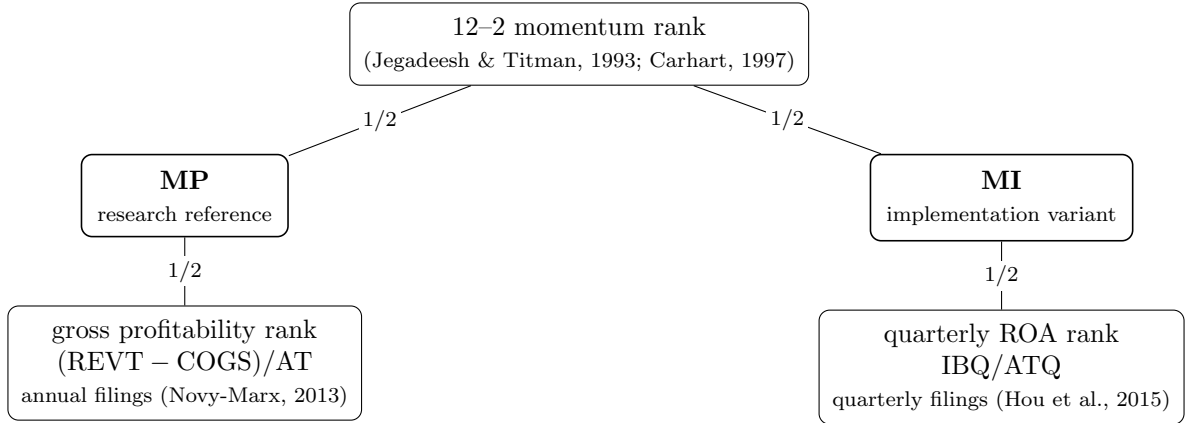
*Research summary of a private codebase (data pipeline, backtest engine, test suite, and research log). Source code and the full chronological research record are available on request. This document describes research output only and is not investment advice.

Universe. At each month-end formation date the universe comprises firms (i) legally incorporated in one of 15 European countries,¹ (ii) with a European primary ordinary listing, and (iii) with company-level market capitalization, aggregated across listings and share classes, between EUR 180M and EUR 1,800M. Financial-format filers (banks, insurers) are excluded via the industrial-format (INDL) filter on fundamentals. The filtered universe averages 927 names per month in sample and 1,024 out of sample.

Returns. Monthly returns are computed from the last available daily close of each month, at the level of the individual security line, with a data-error filter that removes monthly returns outside $[-90\%, +200\%]$. Returns are price-only: the vendor’s total-return adjustment is not of usable quality in this segment, so both strategy and benchmark returns exclude dividends (Section 8).

3 Strategy construction

Two strategies are evaluated, both equal-weighted composites of two cross-sectional percentile ranks:



The two composites share the momentum sleeve and differ in a single slot. MI’s quarterly measure is a ROA-family profitability signal in the spirit of the q -factor literature; its rationale is signal freshness, four filing updates a year against one. The choice between MP and MI was, however, made with out-of-sample results visible (Section 8).

Accruals (Sloan, 1996) and asset growth (Cooper et al., 2008) were also evaluated. Both carry significant Fama–MacBeth coefficients, but their composites underperform the profitability variants in the decile backtest and were not selected.

Timing discipline. The same rules apply to every factor. Accounting data enters factor ranks only from filing availability, proxied as fiscal period end plus 90 days because Compustat Global provides no report dates, and filings drop out of the ranks after a staleness cap. The merge is a backward as-of join, so no filing can enter a formation month before its availability date. Momentum and forward returns use exact calendar-month lookups that treat listing gaps as missing observations. Portfolio selection never conditions on forward return availability. These invariants are enforced by runtime assertions and a test suite (Section 9).

¹GBR, DEU, FRA, NLD, SWE, CHE, ITA, ESP, BEL, DNK, NOR, FIN, AUT, IRL, PRT. Incorporation proxies operating exposure; the mismatch (holding companies, domiciled multinationals) is predominantly a large-cap phenomenon and is small in this size segment.

Strategy	Period	D10	Bench	Excess	t	Sharpe	IR	Max DD	Turnover
MP	IS (2007–19)	+13.4%	+3.5%	+9.0%	5.34	0.92	1.37	−53%	265%
MP	OOS (2020–25)	+12.2%	+2.6%	+8.9%	4.04	0.71	1.31	−26%	261%
MI	IS (2007–19)	+17.3%	+3.5%	+12.5%	6.19	1.10	1.78	−54%	314%
MI	OOS (2020–25)	+18.5%	+2.6%	+14.4%	4.24	0.99	1.90	−27%	314%

Table 1: D10 long-only performance, EUR, price-only returns, net of transaction costs at 30 bps round-trip $\times$ one-sided turnover; the benchmark is a costless paper index. D10 and Bench are annualized compound returns of the portfolio and the equal-weighted factor-eligible benchmark. Excess is the annualized mean monthly return of D10 over Bench; as an arithmetic mean it does not equal the difference of the compound columns. t uses Newey–West standard errors (12 lags); IR is the information ratio versus the same benchmark; turnover is annualized one-sided turnover. Sharpe ratios use a zero risk-free rate (Section 4). The cost deduction amounts to 0.78% (MP) and 0.94% (MI) per year; the break-even round-trip cost that zeroes out-of-sample excess is ≈ 370 bps (MP) and ≈ 490 bps (MI).

4 Backtest design

At each formation month-end, stocks are ranked on the composite, and the top decile of roughly 100 names (D10) is held equal-weighted for one month. The benchmark is the equal-weighted factor-eligible universe, which removes the small-cap and universe-construction tilts from the excess return. The in-sample period covers formations 2007–2019 (156 months) and the out-of-sample period 2020–2025 (71 months). The split date was fixed before evaluation and never moved. Transaction costs are charged at 30 bps round-trip against realized one-sided turnover, and performance figures are net of modeled costs throughout. Attribution regressions (Fama–MacBeth, factor-model alphas) are gross and labeled where they appear: a cross-sectional regression has no portfolio cost concept, and netting a time-series regression would make its estimates depend on the cost assumption. Section 5 reports the resulting cost drag and the break-even cost levels. Significance of mean monthly excess returns uses Newey–West standard errors with 12 lags. Sharpe ratios are computed with a zero risk-free rate throughout. For long-only series, deducting the euro short-term rate (ESTR, averaging 1.3% over this out-of-sample window) would lower the ratio by roughly r_f/σ , about 0.08. Excess returns, their t -statistics, and information ratios are unaffected. Sharpe and information ratios annualize with $\sqrt{12}$ under an i.i.d. assumption. Section 8 quantifies the autocorrelation correction this omits and the sampling error of the ratios.

5 Long-only results

Table 1 reports the headline results. Figures 1 and 2 show the cumulative path and the out-of-sample decile ladder for MP.

Out-of-sample decay relative to in-sample is minimal for MP (9.0% $\rightarrow$ 8.9%) and absent for MI, though Section 8 qualifies how much weight the out-of-sample figures can bear. The decile ladder (Figure 2) is approximately monotone with a strongly negative bottom decile, consistent with a genuine cross-sectional signal.

Factor-model alphas. MP D10 long-only returns, gross of trading costs (Section 4) and converted to USD to match the factor basis, are regressed out of sample on the European Fama–French factors (Fama & French, 2015) plus momentum, both from the Kenneth R. French data library, with Newey–West (4 lags) standard errors. This gives CAPM $\alpha = +4.1\%$ ($t=0.88$), FF3 $\alpha = +9.2\%$ ($t=3.18$), and FF6 $\alpha = +3.0\%$ ($t=1.23$), with loadings concentrated where the construction puts them: SMB +1.19 and UMD +0.44, both $p < 0.01$. The strategy’s return is thus largely compensation through known factor channels deliberately loaded: the FF6 alpha is

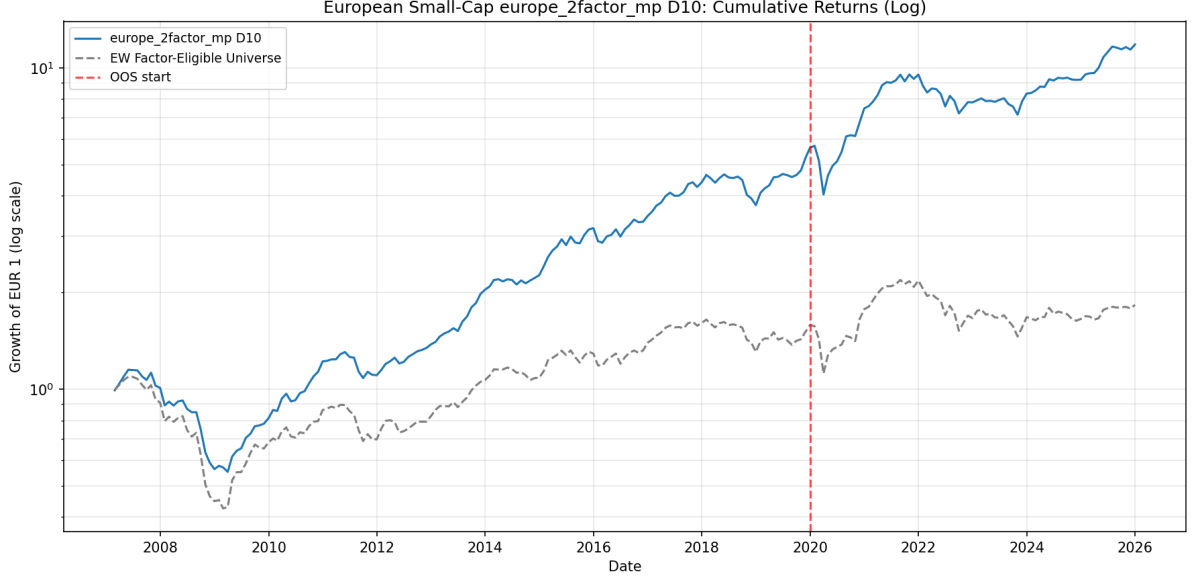


Figure 1: Cumulative growth of EUR 1 (log scale), MP D10 versus the equal-weighted factor-eligible universe, price-only returns, gross of costs. The vertical dashed line marks the out-of-sample split (2020-01), fixed before evaluation.

statistically indistinguishable from zero once a broad-market momentum factor is allowed to absorb the momentum sleeve. What the long-only premium adds over passive factor exposure is decile concentration: D10 holds the top 10% of small caps, while the factor portfolios are built from the top 30% of all stocks.

6 Market-neutral implementation

A tradable market-neutral variant shorts XXSC.DE (Xtrackers MSCI Europe Small Cap UCITS ETF, accumulating share class) against the D10 book. The hedge ratio is a trailing 24-month OLS beta re-estimated monthly on months $[T-24, T-1]$, so no information from the hedged month enters its own hedge ratio. A 50 bps annual borrow cost is charged on the short leg. The long side is price-only while the accumulating ETF's price series is total-return, so results are bracketed by a flat dividend adjustment on the long side of $[0\%, +2\%]$ per year. The lower bound is assumption-free and understates by the long book's actual dividend yield. The upper bound is a conservative estimate of the segment's yield.

Strategy	Net of all modeled costs		CAPM (gross of trading cost)	
	Ann. return	Sharpe	$\beta_{\text{Mkt}} (t)$	$\alpha (t)$
MI hedged	+12.3% – +14.3%	1.44 – 1.68	–0.03 (–0.6)	+13.5% – +15.5% (3.0–3.4)
MP hedged	+6.2% – +8.2%	0.85 – 1.13	–0.02 (–0.4)	+7.2% – +9.2% (2.2–2.8)

Table 2: Hedged D10, out-of-sample (71 months). Newey–West t -statistics in parentheses; for the α column the two t -values correspond to the two ends of the reported range. Ranges span the $[0\%, +2\%]$ dividend adjustment; the hedge ratio is invariant to it. “Net” deducts the 50 bps borrow and long-side trading cost at 30 bps $\times$ turnover. Volatility is 8.5% (MI) and 7.3% (MP) throughout. Sharpe ratios use a zero risk-free rate; the spread is self-funding, so this coincides with the standard excess-return definition. Sharpe ratios annualize with $\sqrt{12}$; the autocorrelation correction this omits is larger for the hedged series than for the long-only series (Section 8). Regressions use Newey–West (4 lags) on the EUR spread.

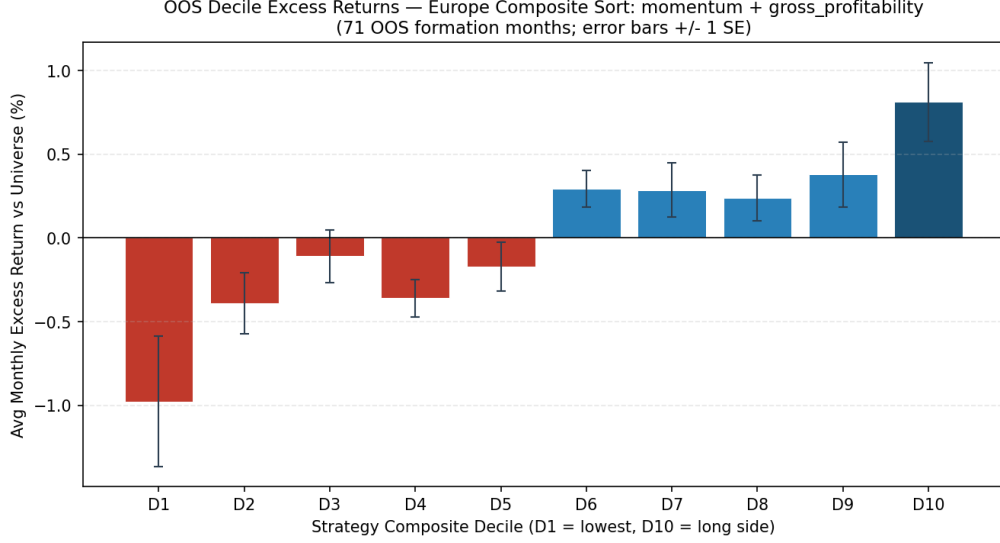


Figure 2: Average monthly gross return by composite decile in excess of the equal-weighted universe, MP, out-of-sample (71 formation months), a signal diagnostic. Error bars are ± 1 standard error of the monthly mean.

Market neutrality is read off the unconditional CAPM beta, which is statistically zero for both strategies out of sample (Table 2). The residual beta against the hedge instrument itself is likewise zero (-0.03 , $t = -0.55$). Neutrality is not factor neutrality: the hedged book intentionally retains small-cap and momentum exposure, with FF6 loadings of $\text{SMB} \approx +0.5$ and $\text{UMD} \approx +0.33$ – 0.42 , both significant. Its FF6 alpha out of sample is $+12.8\%$ – $+14.8\%$ ($t \geq 3.5$) for MI but only $+4.4\%$ – $+6.4\%$ ($t = 1.7$ – 2.5) for MP, so factor-adjusted-alpha claims rest on MI.

The factor returns are USD-based. A funded portfolio carries the EURUSD move at first order, so the long-only returns of Section 5 are converted to USD before regression. The zero-notional spread has no principal to translate: its EUR and USD measurements differ only by a second-order cross term, and it is regressed in EUR. Converting the collateral as well would add its full EURUSD return.

7 Capacity

A separate capacity study replays the strategy under an explicit execution model:

- *Participation limits.* A position in name i is built or unwound at up to a daily participation rate p of the name’s daily traded euro volume V_i , for at most d trading days per rebalance. The per-name weight cap is therefore

$$c_i = \frac{d \cdot p \cdot V_i}{\text{AUM}}.$$

The budget $d \cdot p$ per unit of AUM determines which books are feasible. Halving the participation rate roughly halves each AUM tier.

- *Conservative volume measurement.* The volume V_i entering the caps is the trailing 63-day median of euro volume on the primary listing. Small-cap volume is right-skewed, with a mean around 1.4 times the median, so a mean-based measure admits names on the strength of single block-print days. The median basis is roughly 30% tighter. Volume on secondary venues is ignored, which understates consolidated liquidity by roughly 13% overall. The names that hit their caps are predominantly single-listed, however, so this moves results by at most about 0.1 points per cell.

- *Liquidity floor.* Names are excluded before selection unless they traded on at least 50 of the trailing 63 days, with a mean daily volume of at least EUR 100k and at least 0.05% of market capitalization. The thresholds were fixed upfront on execution-feasibility grounds. The floor removes about 27% of top-decile names, the same share it removes from the full universe, and leaves the excess return unchanged, direct evidence that the edge does not depend on the least liquid names.
- *Allocation.* Where caps bind, the book is market-cap-weighted within per-name caps, and capital a capped name cannot absorb spills to the next names in factor-rank order.
- *Costs and scope.* All trades pay the 30 bps round-trip cost, including forced sales. These arise when a held name drops out of the universe, mostly by appreciating past the market-cap ceiling, or when a position comes to exceed what its diminished volume could liquidate within the exit window. The simulation books these exits at month-end prices. That flattens ceiling exits, because the price move that causes the exit also sets the sale price, and penalizes volume-collapse trims, because the median understates the volume actually available in those names. Together the two effects move results by at most about 0.3 percentage points annualized. Price impact of the trading itself is not modeled: the participation cap is the model’s only impact control, and every figure is pre-slippage.

The signal decays slowly in the holding horizon: pre-cost, a quarterly rebalance retains roughly 87% of the monthly alpha, and a semi-annual one about 70%. The effective lever at scale is therefore the execution budget rather than the rebalance label. The binding constraint is structural rather than per-name illiquidity: alpha concentrates at the smaller end of the 180–1,800M band, so capacity-driven reallocation pushes the book toward larger names where the factor edge is weakest.

Configurations with equal execution budgets admit the same portfolio, but a longer build window at a lower participation rate places a smaller daily footprint in the market for the same book. The recommended operating mode therefore spends the budget as monthly rebalancing with a 120-trading-day build window at 3% daily participation and a 60-trading-day exit ceiling at 10%. Operationally, a name’s per-rebalance entry trade is capped at what 3% of its median daily volume executes in 120 trading days, or $d \cdot p = 3.6$ days of volume. Because that window is several times the month between rebalances, a large position is worked continuously across successive rebalances toward its moving target. The exit ceiling caps the held position itself at what 10% of median daily volume liquidates in 60 trading days, or 6 days of volume, so a position at the ceiling clears in about three monthly rebalances. Names that leave the selection are unwound under the same rate limits. Median positions sit at 3–15% of the exit ceiling. The mode is the best-performing configuration on the tested grid of 90- and 120-day build windows and 30- to 60-day exit ceilings. Configurations outside this grid were not tested.

Table 3 reports the mode’s net excess by AUM tier. Under this execution model, the useful ceiling is near EUR 500M. A quarterly alternative, a 30-day build window at 10% participation with no ceiling on held positions, prints higher paper alpha at the largest tier: MP +6.0% and MI +8.1% at EUR 500M. It buys that by trading at more than three times the daily participation, and it holds 41–46% of NAV above what one month’s volume could liquidate, against 14–17% under the recommended mode. A flat 5% NAV position cap, added for single-name concentration control, costs within measurement noise at every tier except MP at EUR 500M. There it costs about 1.1 points full period and 2.5 out of sample, so a looser cap is indicated.

A square-root argument puts a scale on the unmodeled price impact. Empirically, the cost per euro of a worked order follows

$$\text{cost per euro} \approx Y \cdot \sigma \cdot \sqrt{\frac{Q}{V}},$$

AUM	Execution	MP	MI
$\leq$ EUR 15M	monthly, equal-weighted	+8.2%	+11.2%
EUR 100M	monthly, 120-day build at 3%	+6.1%	+9.8%
EUR 250M	monthly, 120-day build at 3%	+5.4%	+8.0%
EUR 500M	monthly, 120-day build at 3%	+4.4%	+7.4%

Table 3: Annualized net excess return (full period) by AUM tier under the capacity simulation, each tier at its recommended operating mode. The simulation’s execution conventions differ from the backtest of Table 1, so levels are indicative and the gradient across AUM tiers is the result.

with $Y \approx 0.5$, σ the daily volatility, and Q/V the order’s total size in days of volume (Tóth et al., 2011). Applied to a build spread over months, the law brackets the cost, because the impact of earlier trading partially decays while the order is still being worked (Bouchaud et al., 2004). If impact decayed fully between daily slices, every euro traded at 3% participation would pay roughly 15–20 bps one-way at $\sigma \approx 2\%$. If the full build carried its impact to completion, a build at the cap of 3.6 days of volume would pay about 2%, and the median position of 0.2–0.9 days of volume 40–100 bps. Combined with the mode’s roughly 250% annual turnover, the implied drag at EUR 500M is on the order of 0.4 percentage points annualized under fast decay and 1–2 under full memory, with builds worked over weeks to months falling between. The regimes also scale differently: under fast decay the drag is set by the participation rate and is nearly flat across AUM tiers, while under full memory it grows roughly as $\sqrt{\text{AUM}}$, steepening the gradient of Table 3. This is an order-of-magnitude illustration only. Even at the full-memory end, MI retains a clear margin at EUR 500M, while the MP margin there becomes thin.

8 Robustness and limitations

1. **Reuse of the holdout window.** The 2020-01 split was fixed before evaluation and never moved, and selection never conditions on forward returns. The out-of-sample window was, however, evaluated repeatedly across research iterations: strategy variants were compared on out-of-sample statistics, and the out-of-sample impact of data corrections was monitored. Repeated exposure to a fixed holdout is implicit selection. The reported out-of-sample figures are not a one-shot test, and the size of the resulting upward bias cannot be estimated from within the sample. Three things mitigate. The factor set is small and taken from prior literature rather than mined from this data. Data corrections were accepted on correctness and moved results in both directions. And the headline t -statistics (≥ 4.0) survive substantial haircuts. The natural unbiased test is prospective evaluation on data realized after this research.
2. **Price-only returns.** Absolute returns are understated by approximately the segment’s dividend yield, estimated at around 2% per year. Excess returns versus the equal-weighted universe are approximately unaffected, since both sides omit dividends. The hedged results bracket the resulting asymmetry explicitly.
3. **No delisting returns.** Compustat Global provides none, so exits enter as missing forward returns. A stress test classifying all 1,179 permanent exits via vendor deletion-reason codes finds 60% coded as M&A against 1.3% as failures, with exits concentrated in the top decile. The missing returns are therefore more likely to be positive acquisition premia than losses. Imposing -100% on every exit still leaves out-of-sample excess positive ($+5.0\%$, $t=1.70$, gross, evaluated at an earlier pipeline vintage).
4. **Transaction costs are an assumption.** 30 bps round-trip is anchored to quoted European small-cap spreads (15–60 bps) and the academic cost literature (Frazzini et al.,

2015; Novy-Marx & Velikov, 2016), with market impact negligible at personal position sizes. Turnover counts name replacement on the equal-weight book; the smaller rebalancing trades that restore equal weights within continuing positions are not charged. Table 1 therefore reports the break-even costs of roughly 370 bps (MP) and 490 bps (MI), and readers may impose their own assumption.

5. **Descriptive-metric conventions.** Sharpe and IR annualize with $\sqrt{12}$ under an i.i.d. assumption, the standard reporting convention, and no autocorrelation correction (Lo, 2002) is applied to the ratios. The correction would be material: Table 4 reports the adjusted ratios next to the quoted ratios of Tables 1 and 2. The corrected annualization is

$$SR_{\text{ann}} = SR_m \cdot \frac{12}{\sqrt{12 + 2 \sum_{k=1}^{11} (12 - k) \rho_k}},$$

where SR_m is the monthly Sharpe ratio and ρ_k the lag- k autocorrelation of monthly returns. Positive autocorrelation lowers the annualized ratio, and $\rho_k = 0$ recovers $\sqrt{12}$. Long-only monthly returns are positively autocorrelated ($\rho_1 \approx 0.19$ full sample). The effect on the IR is within $\pm 10\%$ and unstable in sign. The hedged series is more sensitive. The persistence enters through the hedge itself: shorting a cap-weighted instrument against an equal-weight universe leaves a residual spread between the two, and that spread is positively autocorrelated. The hedged return decomposes as

$$r_t^{\text{hedged}} = (r_t^P - r_t^B) + (r_t^B - \beta_t r_t^X) - c_t,$$

where r^P , r^B , and r^X are the portfolio, equal-weight benchmark, and hedge-instrument returns, β_t the rolling hedge ratio, and c_t costs. The first bracket is stock selection. The second is that residual spread, and it carries the persistence. In realized terms the persistence means that hedged losses cluster: drawdowns build over sequences of consecutive small monthly losses rather than in single large months. Out-of-sample estimates rest on 71 months and are noisy. The significance tests, including the hedged CAPM alphas, are HAC-based throughout and unaffected. Together with the zero risk-free convention, ratio levels should be compared within this document rather than across publications.

6. **Sampling error of the ratios.** The ratio point estimates carry material estimation error. Estimated over T months, the monthly Sharpe ratio SR_m has asymptotic standard error

$$SE(SR_m) = \sqrt{\frac{1 + SR_m^2/2}{T}}$$

(Lo, 2002); estimate and standard error annualize with the same $\sqrt{12}$. This gives approximately 0.4 over the 71 out-of-sample months and 0.25 over the full sample. Table 4 reports the resulting 95% confidence intervals. The MP out-of-sample Sharpe is within two standard errors of zero. The full-sample rows pool both windows (227 months) and calibrate the wide out-of-sample intervals: under a Sharpe that is stable across the sample, both zero and the upper ends of the out-of-sample intervals are implausible. The excess-return t -statistics test the mean directly with HAC standard errors and already embed their own sampling error.

7. **Country concentration.** Nordic markets contribute roughly 40% of out-of-sample excess on a $\approx 27\%$ universe weight. Alpha sources rotated between sub-periods, from the UK in sample to Sweden out of sample, so this reads as regime exposure rather than a single-market artifact. No country cap is imposed.

8. **Momentum crash exposure.** Half the composite is momentum, which carries documented negative skew and crash risk near market turning points (Daniel & Moskowitz, 2016). The exposure is visible in this strategy’s own record. Realized out-of-sample monthly excess returns are left-skewed, with skewness -0.7 for MP and -0.3 for MI. The worst out-of-sample month for both variants is the November 2020 momentum reversal, at -6.8% and -5.9% excess in a month whose absolute returns were positive. In sample, the profitability sleeve shortened the recovery from the 2009 momentum crash from 57 months for momentum alone to 25 months, mitigating but not removing the exposure.
9. **Conditional risk management.** State-dependent risk overlays on the hedged book were tested and rejected, each failing at a different step.
 - *Hidden regime state.* A two-state Markov-switching description of daily European momentum factor returns loses to a single fat-tailed GARCH- t regime by a BIC margin above 1,200, and the GARCH fit leaves no residual structure in the standardized residuals or their squares. There is no second state to condition on.
 - *Observable volatility state.* Volatility-managed scaling of the hedged spread (Barroso & Santa-Clara, 2015) was tested with estimation windows from six months of monthly returns to the 126-day daily estimator of the original rule, conditioning on the spread’s own volatility and on an in-universe momentum factor’s volatility: no configuration beats constant sizing at matched volatility by more than 0.06 Sharpe in any sample cell. The crash risk such rules de-lever sits in momentum’s conditional market beta, which turns sharply negative after market declines (Daniel & Moskowitz, 2016) and is concentrated in a short loser leg this book does not carry, while the hedge holds the remaining market exposure near zero.
 - *Crowding state.* A comomentum measure (Lou & Polk, 2022), built from residual pairwise correlations within momentum deciles, flags states in which forward hedged returns are on average higher than in calm states: the risk is compensated. A de-risking rule on it avoided all four worst months of 2019–2024 and still gave up 76 percentage points of total return, with the Sharpe ratio falling from 1.56 to 1.02, under an in-sample threshold and zero switching costs. Constant half-sizing at matched volatility keeps the full 1.56, so the rule is dominated by running smaller.

No early-warning variable for the remaining factor-internal drawdowns was validated.

10. **Nonlinearity and interactions.** The equal-weight linear combination was tested against a gradient-boosted alternative under a pre-registered design. A depth-3 LightGBM model was trained on all six factor ranks with yearly walk-forward refits, and its prediction rank was orthogonalized each month against the linear span of the six ranks. The residual carries detectable content: $t=2.4$ over 96 out-of-fold months in sample, same sign at $t=1.1$ out of sample with a roughly unchanged coefficient. A depth decomposition attributes most of it to nonlinear payoff shapes within single factors. A targeted product-term test of the momentum $\times$ profitability interaction reads $t=0.8$. The content is not implementable at meaningful weight: added as a third equal-weight signal, it reduces net out-of-sample excess by 6.5 percentage points per year against a paired standard error of 1.8, since a signal with a standalone decile spread near 1.4% per year displaces selection weight from signals delivering an order of magnitude more. The statistical read rests on a single seed and sits close to its significance threshold. The linear combination is retained on these measured grounds.

Series	Window	N	Annualized Sharpe		Sampling error	
			$\sqrt{12}$	Adjusted	SE	95% CI
MP long-only	full	227	0.84	0.63	0.23	[0.38, 1.30]
MP long-only	OOS	71	0.71	0.62	0.42	[-0.11, 1.52]
MI long-only	full	227	1.06	0.77	0.24	[0.60, 1.52]
MI long-only	OOS	71	0.99	0.81	0.42	[0.17, 1.82]
MP hedged	OOS	71	0.85–1.13	0.73–0.98	0.42	[0.04, 1.96]
MI hedged	OOS	71	1.44–1.68	0.90–1.05	0.43	[0.60, 2.53]

Table 4: Annualized Sharpe ratios with autocorrelation adjustment and sampling error, all net of modeled costs (long-only rows on the basis of Table 1, hedged rows on the net column of Table 2). The two point-estimate columns annualize the same monthly Sharpe ratio with $\sqrt{12}$ and with the autocorrelation-corrected factor. SE and the 95% confidence interval are the asymptotic values for the $\sqrt{12}$ -annualized estimate. The interval does not apply to the Adjusted column, whose sampling error is larger and is not quoted. Hedged ranges span the [0%, +2%] dividend adjustment and the interval spans the bracket. Hedged rows cover the out-of-sample window only: the rolling 24-month beta warmup removes 2008–09 from the hedged sample, so a full-period hedged row would not be comparable to the long-only full period.

9 Software and reproducibility

The pipeline is a Python codebase organized as pull $\rightarrow$ universe $\rightarrow$ factors $\rightarrow$ backtest, with post-hoc diagnostics separated from the core pipeline. Correctness rests on three mechanisms. First, invariant enforcement: point-in-time merges, exact-calendar timing, factor-direction definitions, and currency-basis checks are asserted at runtime and pinned by a test suite of roughly 150 tests, including property-based tests and an end-to-end snapshot regression against committed fixtures. Second, data versioning: WRDS cache snapshots with SHA-256 manifests make any backtest reproducible against a dated data vintage. Third, no silent degradation: missing FX, schema shortfalls, and vendor-data anomalies surface as hard errors or threshold-monitored warnings. These mechanisms have caught consequential errors. In one instance, a month-boundary query that omitted security-level partitioning mixed cross-listing price series non-deterministically, inflating returns by 3–8% per year. The fix is pinned by a query-shape regression test.

References

- Barroso, P., & Santa-Clara, P. (2015). Momentum has its moments. *Journal of Financial Economics*, 116(1), 111–120.
- Bouchaud, J.-P., Gefen, Y., Potters, M., & Wyart, M. (2004). Fluctuations and response in financial markets: the subtle nature of “random” price changes. *Quantitative Finance*, 4(2), 176–190.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52(1), 57–82.
- Cooper, M. J., Gulen, H., & Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4), 1609–1651.
- Daniel, K., & Moskowitz, T. J. (2016). Momentum crashes. *Journal of Financial Economics*, 122(2), 221–247.
- Fama, E. F., & French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.

- Frazzini, A., Israel, R., & Moskowitz, T. J. (2015). Trading costs of asset pricing anomalies. Working paper, AQR Capital Management.
- Hou, K., Xue, C., & Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3), 650–705.
- Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- Lo, A. W. (2002). The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4), 36–52.
- Lou, D., & Polk, C. (2022). Comomentum: Inferring arbitrage activity from return correlations. *Review of Financial Studies*, 35(7), 3272–3302.
- Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1), 1–28.
- Novy-Marx, R., & Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1), 104–147.
- Sloan, R. G. (1996). Do stock prices fully reflect information in accruals and cash flows about future earnings? *The Accounting Review*, 71(3), 289–315.
- Tóth, B., Lempérière, Y., Deremble, C., de Lataillade, J., Kockelkoren, J., & Bouchaud, J.-P. (2011). Anomalous price impact and the critical nature of liquidity in financial markets. *Physical Review X*, 1(2), 021006.